

Assignment 1

Applied Math 3 – 2025

Overview:

For this assignment, choose one of the topics you explored in the first two weeks of class (Input and Movement) and prepare a professional looking report on what you learned.

Alternatively, if both weeks research are related, you could alternatively choose to combine them both into a single report. (You might find this is actually harder than sticking to one topic).

Note: For the next assignments, we will open it up to larger topics potentially covering multiple weeks but will keep the same general format.

Additional Notes:

See the Basic Process & Guidelines for AM3 Classes document for an overview of the research process and some suggested guidelines, timelines and best practices.

Deliverables:

- Summary of research with supporting images, screenshots or other documents (PDF document)
 - Should be 250-500 words and include inline images.
 - Be formatted and written to professional standards including spelling, grammar and layout.
 - I'm happy to have you lay it out in Figma, as long as you are keeping the size down so you can export your work (as a section) in a reasonable way.
- Demonstrated project example that demonstrates and explains what you learned. (PDF document)
 - This can be handed in as a clearly sequenced and explained set of images.
 - Can also provide a short video to help demonstrate the example.

Some topics to make sure you touch on:

1. Questions:

Include your initial 5 questions. Give them a difficulty rank (e.g. Will it take hours/days/weeks to investigate) and highlight the topic you are most interested in exploring.

Include a paragraph explaining your rationale for why this is a useful research topic and why you chose your specific research question to pursue.

2. Initial Research:

Include a paragraph or two and relevant images/video on your initial research, including any useful or key findings.

3. Summarize and explain your findings:

Summarize and explain your initial findings. Include explanation of why you chose a new question or dug deeper into your initial question. Include images or video as needed.

4. Elaborate on your findings:

Include a summary of your secondary findings. How did you build upon your initial research, or why did you change directions should be clearly outlined, as well as what you learned.

5. Demonstrate your understanding:

Create an example that showcases what you learned in a tangible and polished example. It should clearly demonstrate the knowledge you gained and ideally include something functional within UE.

Due Dates:

Friday Sept 12th, 11:59PM

Rubric: [3 Marks Each]

1. Initial Questions - Questions clearly identified, one selected and reasoning explained (e.g. priority, difficulty or some other preference).
2. Initial Research - Includes your findings, links to relevant documents, screenshots demonstrating your exploration in UE, and a brief explanation of your process or hypothesis.
3. Summarize & Evaluate Initial Stage – An evaluation of your initial research, conclusions drawn, and your reasoning explained and justified. Include the follow up question that will guide the next step of your research.
4. Elaborate your findings (Follow Up Question) – Includes your findings, links to relevant documents, screenshots demonstrating your exploration in UE, and a brief explanation of your process or hypothesis.

- *(Note: Same as #2 but for the new question determined in 3.)*

5. Summarize & Evaluate 2 (Follow Up Question) – An evaluation of your follow up research, as well as your follow up question was clearly stated and your reasoning explained and justified.
(Note: Same as #3 but for the follow up question)

6. Project Proposal: How will you demonstrate your knowledge that you have learned? You should include clear guidelines (how will you do it, what are you hoping to find, and how) and include some detail on the steps you will need to take to get there. A rough estimate would be nice to see.

Final Project Evaluation [5 Marks Each]

7. Project Effectiveness: How well does your demonstration of the project clearly communicate the information you set out to show off? It should be clear and concise and include relevant context for full marks.

8. Multimedia Presentation: Images and/or (a single) video is effectively used to show your findings. These should be clearly laid out (e.g. in a PDF) or labeled (a video). I don't want a bunch of random loose images or videos, please present them effectively.
9. Clarity and Communication: The overall package clearly and effectively demonstrates the thesis, process and conclusion you set out to show. Your writing is clear, legible and free of spelling and grammatical errors.
10. Technical Component: This is the UE part of things, and highlights the technical skill and how effectively you were able to use UE to demonstrate your learning.